
Algorithm 1 Meta-evolution

Inputs: baseline; G_{total} total generations; G_{simplify} simplification generations;
 O offspring; P population size; N_{init} initial evaluation seeds;
 N_{reeval} reevaluation seeds; F fitness metric.

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1: population  $\leftarrow$  [baseline]
2: for  $i = 0, \dots, G_{\text{total}} - 1$  do
  # Generate offspring
3:  for  $o = 0, \dots, O - 1$  do
4:    sr_operator  $\leftarrow$  RANDOMCHOICE([mutation, survival, selection, loss])
5:    if  $i = 0$  then
6:      mode  $\leftarrow$  explore
7:    else if  $G_{\text{total}} - i < G_{\text{simplify}}$  then
8:      mode  $\leftarrow$  simplify
9:    else
10:     mode  $\leftarrow$  RANDOMCHOICE([explore, refine, crossover, simplify])
11:   end if
12:   parent  $\leftarrow$  TOURNAMENTSELECTION(population,  $n = 2$ )
13:   if mode = crossover then
14:     parent2  $\leftarrow$  TOURNAMENTSELECTION(population,  $n = 2$ )
15:   end if
16:   feedback  $\leftarrow$  GENERATETRACEFEEDBACK(tasks, parent)
17:   prompt  $\leftarrow$  GENERATEPROMPT(sr_operator, mode, parent, feedback, parent2)
18:   repeat
19:     |  $x \leftarrow$  LLM_ensemble.sample().sample_completion(prompt)
20:   until PASSESVALIDATION( $x$ )
21:   offspring  $\leftarrow x$ 
22: end for
  # Parallel evaluation of offspring and existing population
23:  evaluation_batch  $\leftarrow$   $\{x: N_{\text{init}} \mid x \in \text{offspring}\}$ 
   $\cup \{x: N_{\text{reeval}} - N_{\text{init}} \mid x \in \text{reeval\_individuals}\}$ 
24:  PARALLELEVALUATION(evaluation_batch, tasks,  $F$ )
  # Form archive of scores, averaged over seeds
25:  for  $x \in \text{offspring}$  do
26:    | archive[ $x$ ]  $\leftarrow \frac{1}{N_{\text{init}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
27:  end for
28:  for  $x \in \text{reeval\_individuals}$  do
29:    | archive[ $x$ ]  $\leftarrow \frac{1}{N_{\text{reeval}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
30:  end for
31:  if  $G_{\text{total}} - i < G_{\text{simplify}}$  then # Task-based population
32:    | For each task, add the individual with highest average score on that task to population.
33:    | Fill remaining slots (up to  $P$ ) with individuals selected by highest overall score.
34:  else # LOC versus score population
35:    | Form the Pareto frontier of score versus LOC.
36:    | Fill remaining slots (up to  $P$ ) with individuals selected by highest overall score.
37:  end if
  # Reevaluate surviving offspring for  $N_{\text{reeval}}$  seeds total
38:  reeval_individuals  $\leftarrow \{x \in \text{offspring} \mid x \in \text{new\_population}\}$ 
39: end for
  # Final reevaluation to ameliorate winner's curse
40: for  $x \in \text{population}$  do
41:  | final_eval[ $x$ ]  $\leftarrow \frac{1}{N_{\text{reeval}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
42: end for
43: best_individual  $\leftarrow \arg \max_{x \in \text{population}} \text{final\_eval}[x]$ 
44: return final_eval, best_individual
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